

***In Silico De Novo* Design and Validation of Synthetic RNA Thermoswitches Via Deep Generative Modelling**

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Abstract

Bacteria like *Escherichia coli* (*E. coli*) are utilised as powerful cellular factories for highly valued heterologous chemicals. However, optimising yield demands precise control over the onset of the synthesis. Premature leaky expression initiates a cascade of unwanted mutant *E. coli*, which impairs volumetric yield. RNA thermoswitches (RNATs) offer a robust yet chemically-free solution. The lock, the stem loop structure in a RNAT, sequesters the ribosome binding site (RBS), repressing translation at 37°C, allowing the *E. coli* to populate to maximum biomass prior to induction. This lock then melts at 42°C, triggering production of the targeted chemical.

Initially, Random Forest models trained on a balanced benchmark of 1,198 natural Rfam thermoswitches and 1,198 length and %GC-matched RefSeq 5' UTRs were evaluated to classify functional switches. This proved inadequate as classifiers trained on static 37°C features relied on family-correlated k-mer shortcuts rather than transferable physical rules. The frozen Evolutionary Versatile Architect (EVA) 1.4B causal language model was deployed with taxonomic conditioning to generate *de novo* candidates. Candidates were isolated through an automated multi-stage biophysical triage pipeline and benchmarked against natural and control sequences using continuous thermodynamic ensembles. While the generative model produced structurally diverse, novel topologies with select candidates demonstrating intermediate switching behaviour, the broader cohort exhibited substantial thermodynamic resistance across the induction window. On average, translation initiation exposure remained largely flat, failing to execute the sharp phase transitions observed in natural systems. These findings highlight the need for future generative workflows to incorporate biophysically grounded reinforcement learning and dynamic kinetic modelling to explicitly reward continuous melting transitions.

1.Introduction

1.1 Yield dilemma in Industrial Biomanufacturing

Modern biomanufacturing relies heavily on engineered microbial hosts such as *Escherichia coli* to produce recombinant therapeutics, industrial enzymes, and high-value metabolites (Servant *et al.*, 2000; Neupert *et al.*, 2008). However, heterologous pathway expression imposes a substantial metabolic burden, creating an acute trade-off between cellular biomass accumulation and target molecule synthesis (Zhang *et al.*, 2023). Mitigating this conflict, bioprocess strategies frequently decouple growth from production. Cells are cultivated to high cell density at 37°C. Applying an external stimulus to initiate synthesis (Neupert *et al.*, 2008; Narberhaus, 2010). Conventional chemical inducers (such as IPTG) remain constrained by basal transcriptional leakiness during the uninduced phase, which reduces volumetric productivity by selecting for non-productive escape mutants (Neupert *et al.*, 2008; Hoynes-O'Connor *et al.*, 2015).

Engineered RNA thermoswitches (RNATs) provide a chemically independent alternative (Kortmann and Narberhaus, 2012). A temperature-sensitive hairpin in the 5' UTR sequesters the Shine-Dalgarno RBS at 37°C and releases it at 42°C, permitting ribosomal docking and translation (**Figure 1**; Neupert *et al.*, 2008).

Directly repurposing natural thermoswitches from repositories such as Rfam presents significant operational challenges in industrial fermentation (Neupert *et al.*, 2008; Narberhaus, 2010). Under environmental stress, natural systems evolved to adjust smoothly like dimmer switches rather than tight binary on/off switches (Kortmann and Narberhaus, 2012; Loh *et al.*, 2018). This continuous, broad unravelling leads to baseline expression leakage at 37°C alongside incomplete activation upon heating (Neupert *et al.*, 2008). Computational *de novo* design provides a route to generate synthetic regulatory elements optimised specifically for biomanufacturing that are strongly repressed during biomass accumulation, undergoing sharp cooperative unzipping at 42°C, and readily adapted to non-native expression cassettes (Neupert *et al.*, 2008; Groher *et al.*, 2019).

RNA thermoswitch mechanism — RBS sequestration vs thermal release

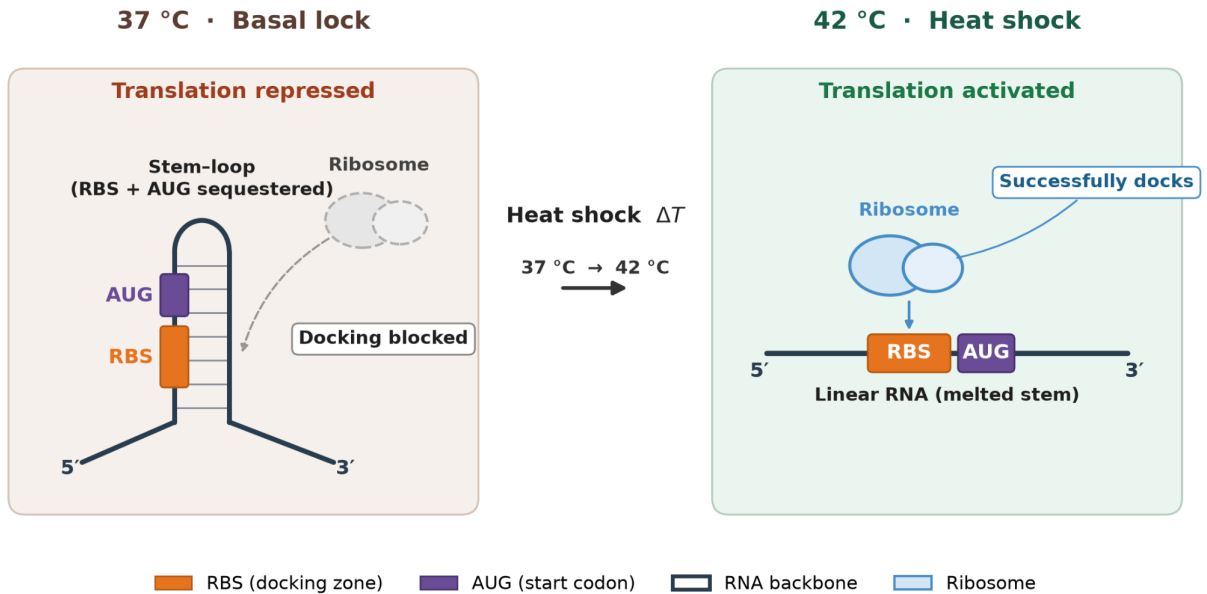


Figure 1: RNA thermoswitch Mechanism: At 37°C, the RNA strand folds into a stable hairpin structure that sequesters the Ribosome Binding Site (RBS), creating a physical lock that represses translation. Upon a 42°C heat shock, the stem-loop melts into a linear conformation, exposing the RBS for ribosomal docking and initiating chemical synthesis

1.2 Computational Bottleneck: Limitations of Predictive Classification

Computational RNA-switch design has moved from constraint-programming and multi-objective inverse folding (Taneda, 2011; Garcia-Martin *et al.*, 2016) to classifiers trained on sequence grammar and thermodynamic descriptors (Groher *et al.*, 2019; Chu *et al.*, 2024; Lindsey, 2026). These methods scale beyond low-throughput screening (Runge *et al.*, 2024), but remain evaluative rather than generative.

Developing models that capture dynamic thermal regulation remains an active computational challenge (Runge *et al.*, 2024; Huang *et al.*, 2026). Unlike static structural RNAs that fold into a single rigid minimum free energy (MFE) state, functional thermoswitches operate across a dynamic thermodynamic ensemble that undergoes a continuous, cooperative transition over a defined thermal window (Neupert *et al.*, 2008; Kortmann and Narberhaus, 2012). Many traditional machine learning setups rely on static, single-temperature scalar representations (such as

baseline folding energies at 37°C), which compress conformational ensembles into discrete snapshot descriptors. While such features capture baseline stability, modelling dynamic switching behaviours such as the continuous melting trajectory, activation threshold (T_m), and cooperativity (h) often requires evaluating continuous thermodynamic partition functions across multi-state ensembles (Runge *et al.*, 2024). Consequently, synthetic biology has increasingly shifted towards generative architectures and biophysical scoring pipelines capable of designing and verifying dynamic RNA systems.

1.3 De Novo design and Validation

We deployed the Evolutionary Versatile Architect (EVA), a frozen 1.4-billion-parameter causal language model, to sample candidate 5' UTRs conditioned on host taxonomy and RNA class (Huang *et al.*, 2026). Downstream ensemble scoring, automated biophysical triage, and Hill parameterisation, are used to filter and characterise those sequences thoroughly (Lorenz *et al.*, 2011; Fornace *et al.*, 2025). The pairing is an end-to-end *in silico* route from unconstrained sequence space to annotated thermoswitch candidates.

2. Data and Tools Used

2.1 Data

Evaluating supervised classification and benchmark *de novo* design, we curated two distinct datasets: an established positive RNAT set, a balanced negative control set (Table 1).

Labelled Rfam / RefSeq panel for supervised RF training					
Cohort	Source	Size (N)	Target Label	Feature Representation	Role in Workflow
Positive RNATs	Rfam	1,198	Positive (1)	Static Tabular Features	Supervised RF Training
Negative Controls	NCBI RefSeq	1,198	Negative (0)	Static Tabular Features	Supervised RF Training

Table 1: Dataset composition for the supervised Random Forest Model

2.1.1 Positive Class: Rfam Database

The positive dataset comprises 1,198 non-redundant prokaryotic RNAT sequences extracted from the Rfam database (Kalvari *et al.*, 2021) and we clustered at 80% sequence identity using CD-HIT (Fu *et al.*, 2012). Rfam organises functional RNA families using Covariance Models (CMs) (Eddy and Durbin, 1994) which extend profile Hidden Markov Models (Eddy, 2011) and we searched using the Infernal software suite (Nawrocki and Eddy, 2013). The harvested cohort spans 35 covariance families, predominantly represented by the *cspA* family (RF01766, ~80% of positive records), with additional representation from *ROSE*, *FourU*, and *prfA* motifs (Kortmann and Narberhaus, 2012; Loh *et al.*, 2018).

2.1.2 Negative Class: RefSeq Background Controls

Supervised binary classification requires an equally representative negative control set. A negative cohort of 1,198 non-switching bacterial 5' UTR windows were sampled from the NCBI RefSeq database (O'Leary *et al.*, 2016), representing a standard non-redundant transcriptomic baseline. We prevented homology data leakage and taxonomic bias by making sure candidate 5' UTRs were decontaminated against all Rfam covariance models using Infernal `cmscan --cut_ga` (Nawrocki and Eddy, 2013). The remaining sequences were paired 1:1 to the positive class via Hungarian bipartite matching on sequence length (spanning 46 to 512 nucleotides) and GC content ($|\Delta\text{GC}| \leq 0.05$) (Kuhn, 1955; Trotta, 2014), establishing a balanced benchmark panel of $N = 2,396$ transcripts (**Table 1**). This matching protocol was designed so that downstream classification performance reflects genuine structural and thermodynamic features rather than baseline compositional artefacts.

2.2 Tools

Category	Package / Tool	Version / Source	Primary Function in Pipeline
Runtime & Data	Python	3.12.5	Base runtime environment and script execution
Runtime & Data	Pandas / NumPy	3.0.5 / Standard	Tabular feature matrix construction and array operations
Runtime & Data	BioPython	Standard	FASTA sequence parsing and sequence I/O management
Machine Learning	Scikit-Learn	1.9.0 (Pedregosa et al., 2011)	Random Forest training, bagging, and cross-validation
Machine Learning	SciPy	1.18.0 (Virtanen et al., 2020)	Non-linear least-squares Hill curve fitting (curve_fit)
Bioinformatics	CD-HIT	Fu et al. (2012)	Sequence redundancy reduction at 80% identity threshold
Bioinformatics	Infernal (cmscan)	Nawrocki & Eddy (2013)	Rfam covariance model search and negative decontamination
Bioinformatics	BLAST+ (blastn)	Camacho et al. (2009)	Nucleotide local alignment for EVA sequence novelty filter
Bioinformatics	HMMER (nhmmer)	Wheeler & Eddy (2013)	Profile HMM sequence alignment against Rfam database
Biophysics	ViennaRNA	2.0 (Lorenz et al., 2011)	Static MFE, Z-score, and 30–60°C partition function sweeps
Biophysics	NUPACK	Fornace et al. (2025)	Static 37°C ensemble free energy and pairing matrices
Generative Model	EVA (GENTEL-Lab)	1.4B CLM (Huang et al., 2026)	Frozen autoregressive de novo 5' UTR generation
Generative Model	PyTorch / CUDA	Official framework	GPU tensor acceleration for EVA inference

Table 2 Summary of core computational software, libraries, and biophysical engines: primary computational packages, versions, and simulation engines utilised across data gathering, predictive machine learning, generative sampling, and *in silico* biophysical validation. For the extended software stack, and background infrastructure specifications, refer to Supplementary Materials.

3. Methods

3.1 Pipeline Architecture

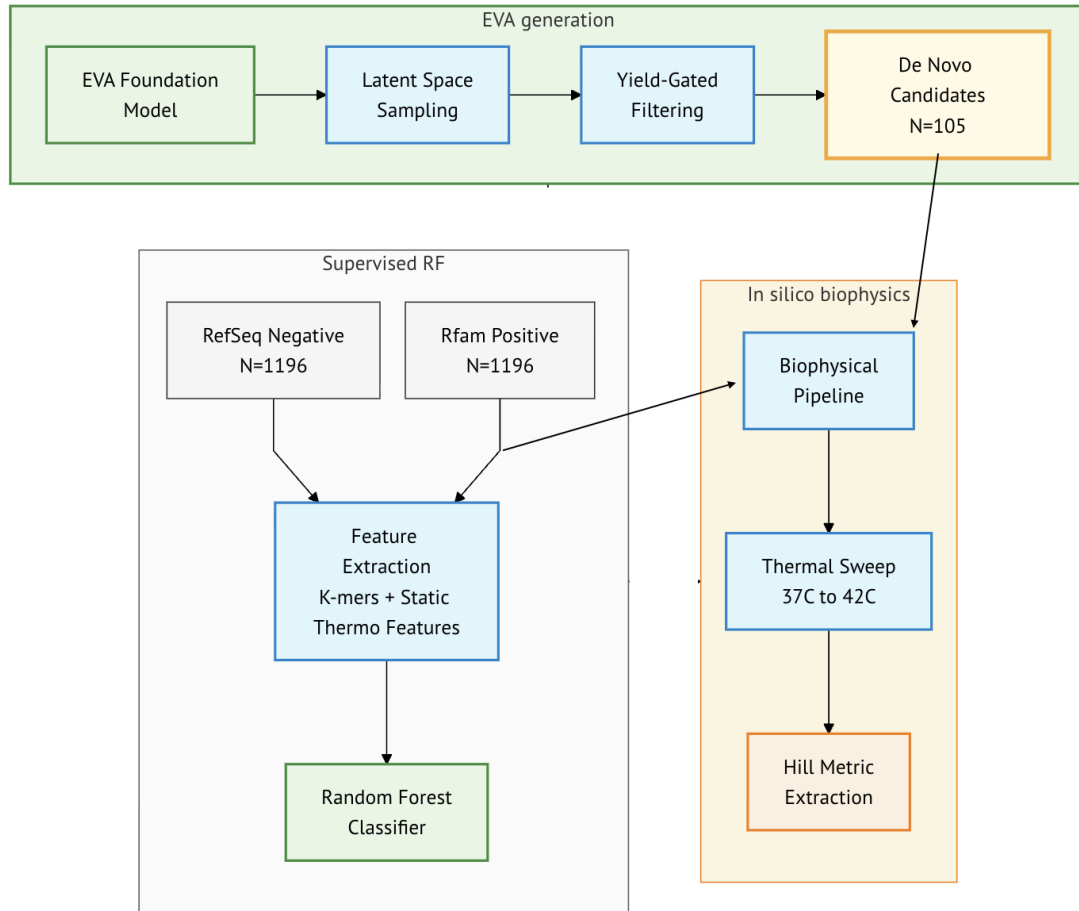


Figure 2: Flowchart of the computational architecture. The flowchart displays three grouped domains: EVA generation, Supervised RF, and *In silico* biophysics and the overall architecture of *de novo* and comparison metrics calculated mainly via NUPACK (Fornace *et al.*, 2025) and ViennaRNA (Lorenz *et al.*, 2011).

3.2 Supervised Predictive Baseline: Random Forest Classification

We utilised the balanced, length- and GC-matched cohort established in Section 2.1 (**Table 1**), the sequences were transformed into a standardised numerical matrix to train the supervised classification baseline.

3.2.1 Sequence and Thermodynamic Feature Extraction and Normalisation

The core of evaluating RNA structural viability is translating a linear nucleotide string into physical and evolutionary parameters (Eddy and Durbin, 1994; Kortmann and Narberhaus, 2012). Sequences were parsed into two distinct feature domains:

K-mer Frequencies: Captures the sequence grammar and family-conserved motifs without sequence alignment, exhaustive sliding-window counts of overlapping dinucleotides ($k = 2$, spanning all $4^2 = 16$ permutations) and trinucleotides ($k = 3$, spanning all $4^3 = 64$ permutations, spanning AAA, AAC, ..., UUU). These were extracted for each transcript using a spectrum kernel formulation (Leslie *et al.*, 2001). To prevent sequence length from confounding count magnitudes, each k-mer count was normalised by total available windows ($N - k + 1$), yielding an 80-dimensional continuous intensive frequency vector. These k-mer profiles allow downstream trees to evaluate localised base-stacking tendencies and evolutionary signatures conserved across natural RNA families.

Static Thermodynamic Scalars: Sequence grammar alone does not dictate physical folding. We mapped baseline structural architecture, through sequence validation evaluated at a static physiological temperature of 37°C utilising the ViennaRNA (Lorenz *et al.*, 2011) and NUPACK (Fornace *et al.*, 2025) software suites. We extracted parameters included ensemble free energy ($\Delta G_{\text{ensemble}}$) and MFE (ΔG_{MFE}), because longer transcripts naturally form more base pairs, raw ΔG_{MFE} scales inversely with sequence length. Length-normalising ΔG_{MFE} into thermodynamic density ($\Delta G_{\text{MFE}}/N$) prevented the classifier from using sequence length as a shortcut.:

$$\Delta G_{\text{density}} = \frac{\Delta G_{\text{MFE}}}{N}$$

where N represents sequence length in nucleotides (Trotta, 2014). This formulation expresses thermodynamic stability as an intensive property independent of transcript length.

3.2.2 Random Forest Classification

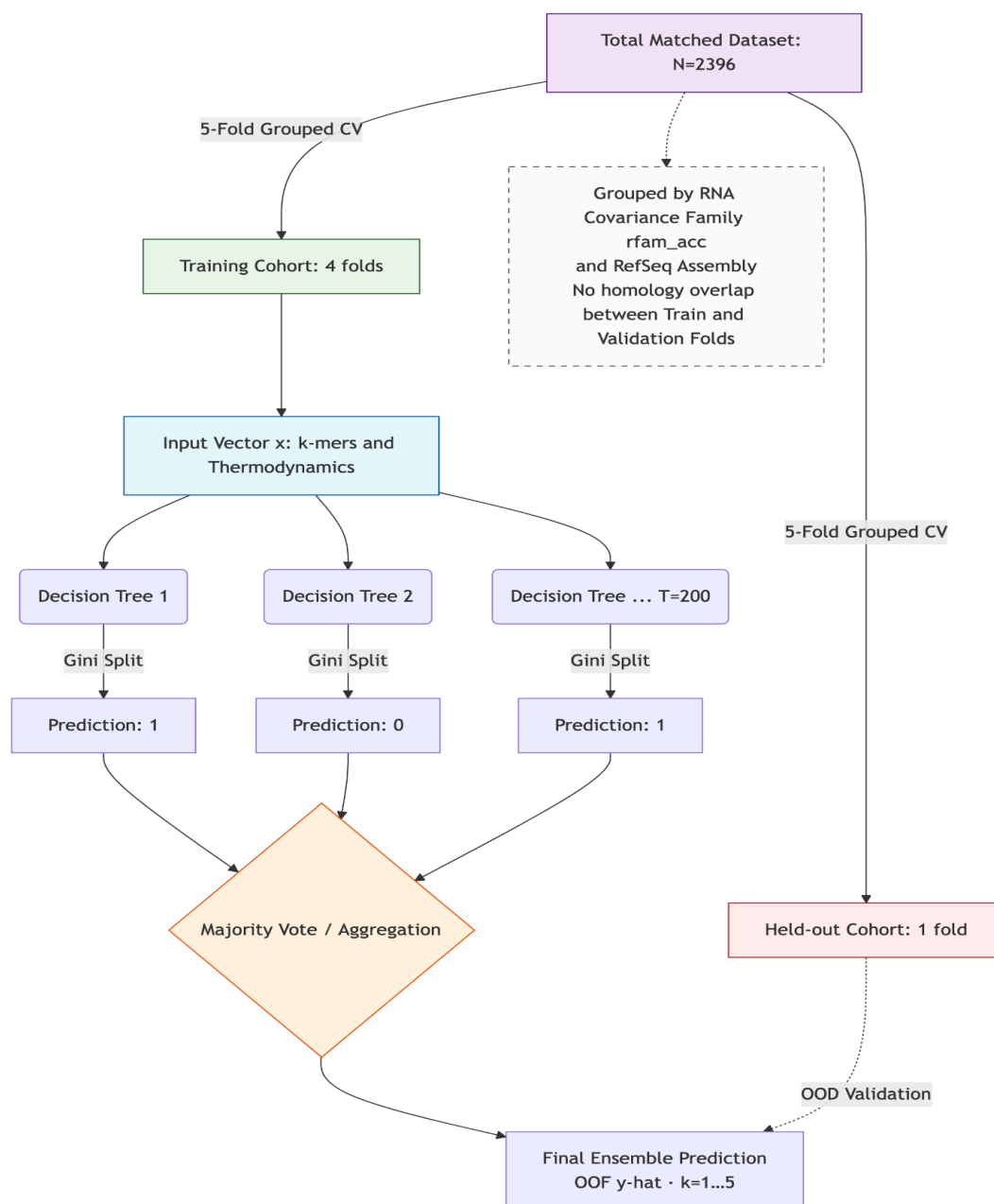


Figure 3 Multi-tiered feature representation pipeline for supervised thermoswitch classification ($D = 92$): Schematic breakdown showing the extraction of sequence composition (mono-, di-, and trinucleotide frequencies; $D = 84$), macroscopic composition ($D = 2$), length ($D = 1$), and static 37°C thermodynamic scalars ($D = 5$) including length-normalised free energy density ($\Delta G_{\text{density}}$). Abbreviations: OOD, out-of-distribution validations; OOF = out-of-fold predictions

We established a predictive baseline by utilising a Random Forest (RF) classifier via scikit-learn (Pedregosa *et al.*, 2011). The RF algorithm constructs an ensemble of decision trees utilising bootstrap aggregating (bagging). For each tree, a random

sample of the training instances is drawn with replacement, ensuring that individual base estimators remain functionally decoupled and solid against dataset variance.

We decorrelated the trees, by letting the model employ random subspace feature bagging. At each candidate split, only a random subset of the 92 input features (comprising 80 k-mer frequencies, static thermodynamic scalars from ViennaRNA and NUPACK, SD-AUG spacing, and compositional metrics) is evaluated. The optimal split is determined by minimising the Gini impurity (G), which quantifies the probability of misclassifying a randomly chosen sequence if it were labelled according to the class distribution at that node:

$$G = 1 - \sum_{i=1}^C p_i^2$$

For a given RNA transcript represented by feature vector $x \in \mathbb{R}^{92}$, the final architecture outputs an aggregated class probability $P(\text{Thermoswitch} \mid x)$ across the ensemble of $T = 200$ decision trees:

$$P(\text{Thermoswitch} \mid x) = \frac{1}{T} \sum_{t=1}^T h_t(x)$$

where $h_t(x)$ represents the probabilistic prediction of the t -th tree. The ensemble operated with unconstrained maximum depth and a fixed random state of 42.

Model performance was evaluated using 5-fold stratified group cross-validation (StratifiedGroupKFold; Pedregosa *et al.*, 2011). Grouping was enforced at the RNA covariance family accession level (rfam_acc) and RefSeq assembly level to ensure that homologous sequence clusters such as those across *cspA*, *ROSE*, *FourU*, and *prfA* motifs (Kortmann and Narberhaus, 2012; Loh *et al.*, 2018) remained partitioned within individual folds. We explicitly did this to prevent sequence homology data leakage between training and validation splits while maintaining precise class equilibrium. The model's predictive capacity was benchmarked on genuinely unseen biological families utilising solely static, 37°C snapshot features.

3.3 Deep Generative Design: EVA *De Novo* design

While supervised classification provides a structured setup for scoring known sequence motifs, standard discriminative classifiers are primarily evaluative rather than generative in scope (Groher *et al.*, 2019; Lindsey, 2026). They are typically configured to evaluate pre-existing transcripts within defined feature boundaries, making the direct construction of non-natural nucleotide backbones less accessible without iterative search heuristics (Taneda, 2011; Garcia-Martin *et al.*, 2016). Reliance on discrete, single-temperature feature representations may limit how effectively an algorithm captures the broad combinatorial landscape needed to design synthetic regulatory RNAs (Runge *et al.*, 2024).

To explore wider sequence spaces apart from natural alignment matrices, we made the workflow incorporate de novo sequence generation via the Evolutionary Versatile Architect (EVA) foundation model (Figure 2; Huang *et al.*, 2026). EVA operates as a frozen, 1.4-billion parameter autoregressive causal language model pre-trained across diverse transcriptomic corpora (Huang *et al.*, 2026). Conditioned on bacterial host syntax and structural RNA classes, EVA was deployed to sample unaligned candidate 5' UTRs, which were subsequently passed to an automated biophysical triage and thermodynamic simulation procedure (Figure 2; Zhao *et al.*, 2024).

3.3.1 Cloud Infrastructure

EVA bypasses traditional biological sequence alignment by framing RNA generation as an autoregressive causal language modelling task conditioned on structural RNA-type and host taxonomic tags (Huang *et al.*, 2026). Because inference across a 1.4-billion parameter neural network requires dedicated hardware acceleration, we deployed pre-trained model weights from the GENTEL-Lab repository to the remote GPU virtual machines running PyTorch and CUDA. Sequence generation runs were executed using the official inference framework and transferred to local environments for automated biophysical triage.

3.3.2 Autoregressive Generation

Sequences were generated autoregressively from the 5' to 3' terminus, where the conditional probability of emitting nucleotide x_i at position i depends on all preceding context tokens ($x_1, \dots, x_{i-1}$) conditioned on functional prefix prompts (c). In this

workflow, conditioning was enforced by passing `--rna_type mRNA` and the *E. coli* taxonomic identifier (`--taxid 562`):

$$P(S) = \prod_{i=1}^L P(x_i \mid x_1, \dots, x_{i-1}, c)$$

This probabilistic sampling allows the generative model to explore regions of sequence space exempt from natural alignment matrices while maintaining basal biological syntax (Huang *et al.*, 2026).

3.3.3 Yield-Gated Triage and Candidate Isolation

Unconstrained generation from deep causal language models frequently yields non-viable or repetitive sequences (Huang *et al.*, 2026). The Primary sampling criteria we used followed: candidate transcripts that terminated at an end-of-sequence (`<eos>`) token within the valid RNA vocabulary (`{A, C, G, U}`) and fell within a 40- to 600-nucleotide length boundary.

Following initial sampling, we applied a soft-drop filtering protocol to eliminate low-complexity sequence collapse (Wootton and Federhen, 1993; Morgulis *et al.*, 2006; Huang *et al.*, 2026). Specifically, candidates were discarded if they exceeded a combined mono- or dinucleotide fraction of 0.85 or exhibited a unique 3-mer ratio below 0.05 (Wootton and Federhen, 1993).

Raw generative output passed through three successive filters: thermodynamic stability ($Z \leq -2$), dynamic unzipping ($\Delta P_{\text{RBS}} > 0$), and sequence novelty ($E_{\text{Rfam}} > 10^{-3}$).

- Thermodynamic Stability ($Z \leq -2$): To verify structural stability beyond composition expectation, an MFE Z -score was evaluated as $Z = (\Delta G_{\text{MFE}} - \mu_{\text{shuf}}) / \sigma_{\text{shuf}}$, where μ_{shuf} and σ_{shuf} represent the mean and standard deviation of MFEs across 100 dinucleotide-preserving sequence shuffles (Trotta, 2014).
- Dynamic Unzipping Potential ($\Delta P_{\text{RBS}} > 0$): Candidates were filtered for positive thermal responsiveness by requiring $\Delta P_{\text{RBS}} = P_{\text{open}}(55^\circ\text{C}) - P_{\text{open}}(37^\circ\text{C}) > 0$.
- Sequence Novelty ($E_{\text{Rfam}} > 10^{-3}$): To ensure divergence from natural families, candidate alignments against the complete Rfam database were evaluated via local alignment and profile hidden Markov models, enforcing $E_{\text{Rfam}} = \min(E_{\text{blastn}}, E_{\text{nhmmer}}) > 10^{-3}$ (Altschul *et al.*, 1990; Wheeler and Eddy, 2013; Zhao *et al.*, 2024).

ViennaRNA (Lorenz *et al.*, 2011) was designated as the primary thermodynamic simulation engine across the triage workflow as it prevented cross-software calibration artefacts during multi-stage filtering.

3.4 *In Silico* Biophysical Validation and Hill Function Modelling

Overcoming the static limitations of the predictive baseline and rigorously evaluating *de novo* sequences generated by the EVA model, a dynamic *in silico* validation pipeline was engineered. We attempted to map the continuous thermodynamic trajectory of each candidate, simulating the precise biological heat-shock window required for industrial biomanufacturing.

3.4.1 Dynamic Thermal Sweep Simulation

Unlike standard classifiers that evaluate a single MFE ground state, functional RNA thermoswitches operate via temperature-dependent shifts across conformational ensembles. We captured this dynamic behaviour. Candidate sequences and Rfam control families were evaluated across a continuous thermal sweep (30°C to 60°C), with the physiological sub-window (37°C to 42°C) designated for induction benchmarking (Neupert *et al.*, 2008).

Dynamic ribosome binding site (RBS) accessibility was parameterised from unconstrained ensemble base-pairing probability matrices $P(T)$ computed via ViennaRNA partition

function algorithms (McCaskill, 1990; Lorenz *et al.*, 2011). For each nucleotide position i , the marginal unpairing probability $u_i(T)$ was calculated as:

$$u_i(T) = 1 - \sum_{j \neq i} p_{i,j}(T)$$

The overall temperature-dependent accessibility feature $A(T)$ was then derived as the arithmetic mean of these marginal probabilities across the 30-nucleotide translation initiation window:

$$A(T) = \frac{1}{|W_{\text{RBS}}|} \sum_{i \in W_{\text{RBS}}} u_i(T)$$

where W_{RBS} denotes the index set of the initiation window. This formulation provides a computationally scalable proxy for local conformational opening, capturing thermal melting trajectories for subsequent non-linear Hill equation fitting without the overhead of computing joint multi-base exposure ensembles.

3.4.2 Hill Function Modelling and Metric Extraction

Parameterising the continuous melting trajectories with an adapted Hill equation allowed direct numerical comparison against natural controls. The transition of an RNA thermoswitch from a locked state to an open state exhibits cooperative melting behaviour (Kortmann and Narberhaus, 2012), which is optimally modelled using an adapted Hill equation (Hill, 1910; Neupert *et al.*, 2008)..

We used non-linear least squares optimisation. Each sequence's thermal trajectory was fitted to the following logistic Hill function:

$$y(T) = y_{\min} + \frac{y_{\max} - y_{\min}}{1 + \left(\frac{T_m}{T}\right)^h}$$

where:

- $y(T)$ is the probability of the RBS being unpaired at temperature T .

- y_{min} and y_{max} represent the baseline structural leakiness (basal state at 37°C) and the maximum unzipping capacity, respectively.
- T_m (Activation Threshold) is the critical melting temperature at which the switch achieves 50% of its maximum conformational change.
- h (Hill Coefficient) quantifies the cooperativity and steepness of the structural transition.

We ensured computational stability across highly variable thermodynamic ensembles, by making sure the fit was executed using `scipy.optimize.curve_fit` within the SciPy ecosystem (Virtanen *et al.*, 2020). Because data-driven box bounds were established-specifically capping $T_m \in [5, 95]$ and $h \in [0.1, 20]$ -the algorithm defaulted to the Trust Region Reflective (TRF) optimisation method (Branch, Coleman and Li, 1999), capped at 10,000 maximum function evaluations.

3.4.3 Validation Criteria

We extracted h and T_m , as the architecture parameterises continuous structural transitions into quantitative biophysical metrics for comparative evaluation. For stable application in bacterial expression systems, a candidate *de novo* thermoswitch should ideally display an activation threshold (T_m) aligned near the targeted 42°C heat-shock window, accompanied by a high Hill coefficient (h) reflecting a cooperative transition rather than a broad, gradual unravelling (Neupert *et al.*, 2008; Kortmann and Narberhaus, 2012). While heat-repressible regulatory systems (such as temperature-sensitive RNase E elements; Hoynes-O'Connor *et al.*, 2015) illustrate the general principles of thermally responsive RNA restructuring, their inverted repression mechanics differ from the translational activation ("ON-switch") profile targeted in this setup. These extracted parameters establish a standardised benchmark we used to evaluate the thermodynamic switching performance of *de novo* generations alongside natural positive controls in the subsequent results analysis.

4. Results

4.1 Random Forest results

4.1.1 Cohort Matching and Control Baselines

Model	Features	Stratified ROC-AUC	Accuracy	vs chance (0.50)
Length-alone RF	seq_length	0.196	0.273	Below chance
%GC-alone RF	%GC	0.254	0.279	Below chance
Length + %GC RF	length + %GC	0.157	0.234	Below chance
Random chance (balanced)	—	0.500	0.500	—

Table 3 Predictive performance of Random Forest models trained on isolated compositional features: across the matched benchmark cohort ($N = 2,396$). Cross-validated scores reflect 5-fold stratified holdout. Sub-random length performance ($\text{ROC-AUC} < 0.500$) represents an expected artefact of 1:1 paired matching under finite-sample partitioning rather than an anti-predictive biological signature (Fawcett, 2006).

We later verified that the negative cohort pairing confirmed that compositional matching successfully eliminated trivial length and %GC separation shortcuts (**Table 3; Figure 4**). Univariate Mann-Whitney analysis also showed no standalone discriminative signal for sequence length, while length-only decision tree ensembles exhibited sub-random performance attributable to discrete 1:1 pair splitting across cross-validation folds (**Table 3**). Overlaid density distributions were found to demonstrate aligned parameter profiles between cohorts (**Figure 4**), ensuring that subsequent classification evaluates structural and thermodynamic signatures rather than primary compositional discrepancies.

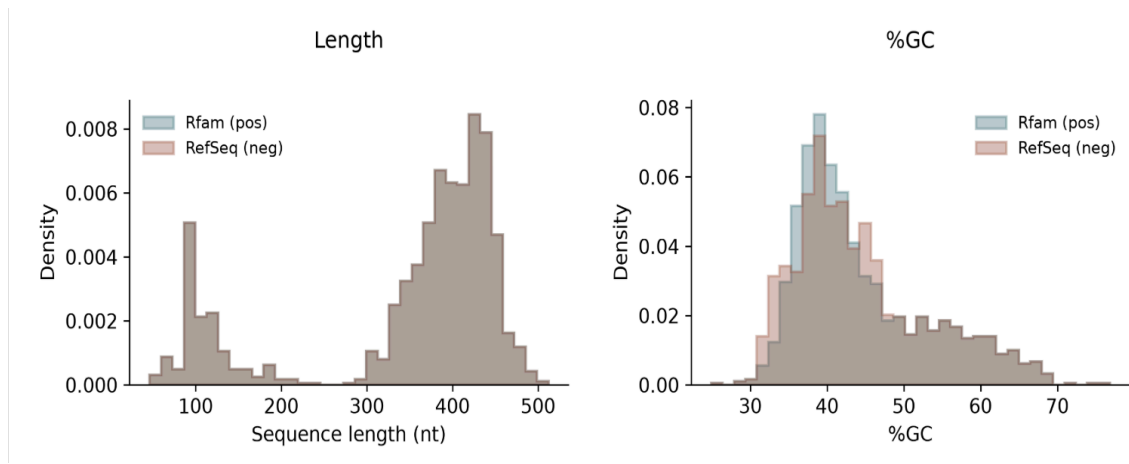


Figure 4 Overlaid probability density histograms of matched sequence cohorts: distributional overlap of sequence length (nt) and percentage GC content (%GC) between the Rfam positive thermoswitches and the RefSeq negative controls

4.1.2 Supervised Baseline: Cross-Validation and Group Holdout Collapse

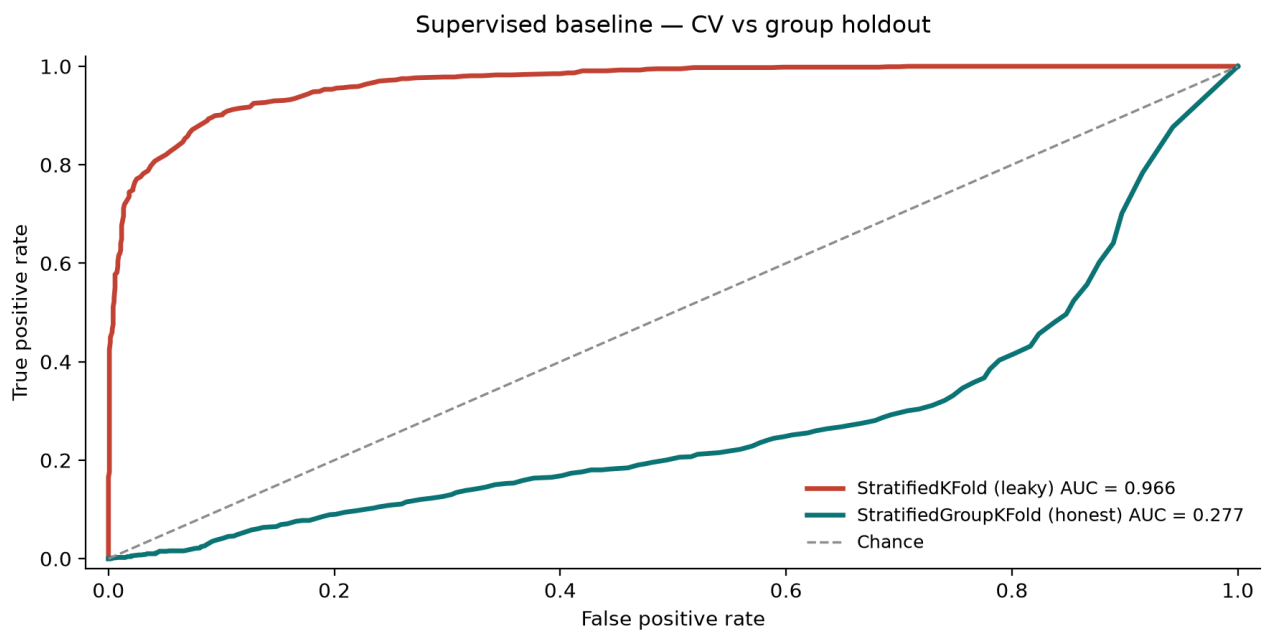


Figure 5 Receiver Operating Characteristic (ROC) curves evaluating cross-validation protocols: This is detailing the collapse of the predictive performance by showing the contrast between standard StratifiedKFold protocol (AUC = 0.966) against the homology-isolated StratifiedGroupKFold (AUC = 0.277)

When we trained and evaluated across standard 5-fold stratified cross-validation, the Random Forest model achieved an apparent near-perfect classification performance ($\text{ROC-AUC} = 0.966$; **Figure 5**). Isolating homologous families inside single folds reversed this result. Grouped cross-validation (`StratifiedGroupKFold`) dropped discrimination to $\text{ROC-AUC} = 0.277$ (**Figure 5**). This failure demonstrates to us that standard random splitting suffers from severe homology leakage, and the high baseline performance was driven by family-specific sequence memorisation rather than transferable biophysical rules governing RNA thermal switching.

4.1.3 Feature Attribution and the K-mer Sequence Memorisation

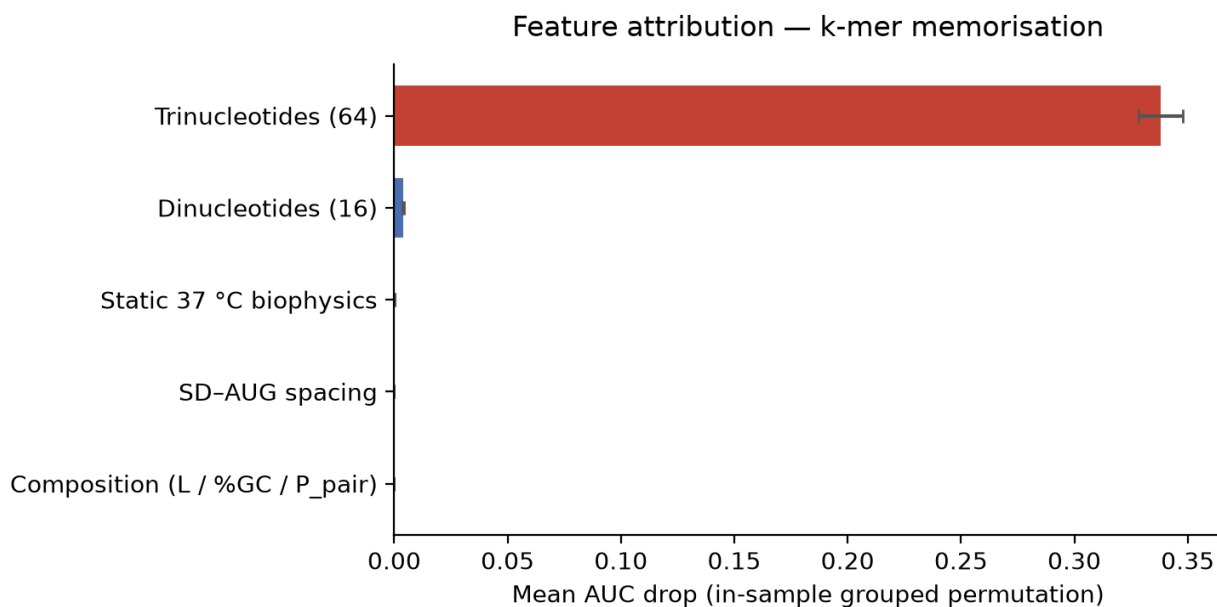


Figure 6 Grouped permutation importance of the Random Forest classifier: Mean ROC-AUC drop occurred when specific features were shuffled and permuted out revealing the dominance of the trinucleotide feature set and the little to no effect of the other features including Static 37°C biophysics which incl. $\Delta\text{GMFE}/N$.

Establishing the basis for this cross-validation collapse, grouped permutation feature importance was calculated from the fitted ensemble. The attribution profile reveals that the discriminative signal was overwhelmingly concentrated within the 64-dimensional trinucleotide frequency block, driving a mean ROC-AUC drop of 0.338.

In contrast, the static 37°C biophysical and compositional blocks contributed negligibly (mean $\Delta\text{AUC} \approx 0$). This is an example of the sequence-memorisation trap. Rather than learning transferable thermodynamic principles of RNA secondary structure stability, the decision trees prioritised family-conserved trinucleotide motifs to separate the classes.

4.1.4 Confidence Calibration and Out-of-Distribution Transfer

Because the model memorises traits specific to certain data families, it exhibits high uncertainty when evaluating unfamiliar, out-of-distribution transcripts. (Figure 7). Evaluating out-of-fold predicted class probabilities ($\hat{y}$) under StratifiedGroupKFold revealed that predictions were heavily skewed towards the low-confidence bin ($\hat{y} \leq 0.20$), while high-confidence assignments ($\hat{y} \geq 0.80$) were exceptionally rare. While the tree ensemble can serve as a localised heuristic for known motifs, static scalar representations at 37°C offer limited generalisability across novel, uncharacterised non-coding families.

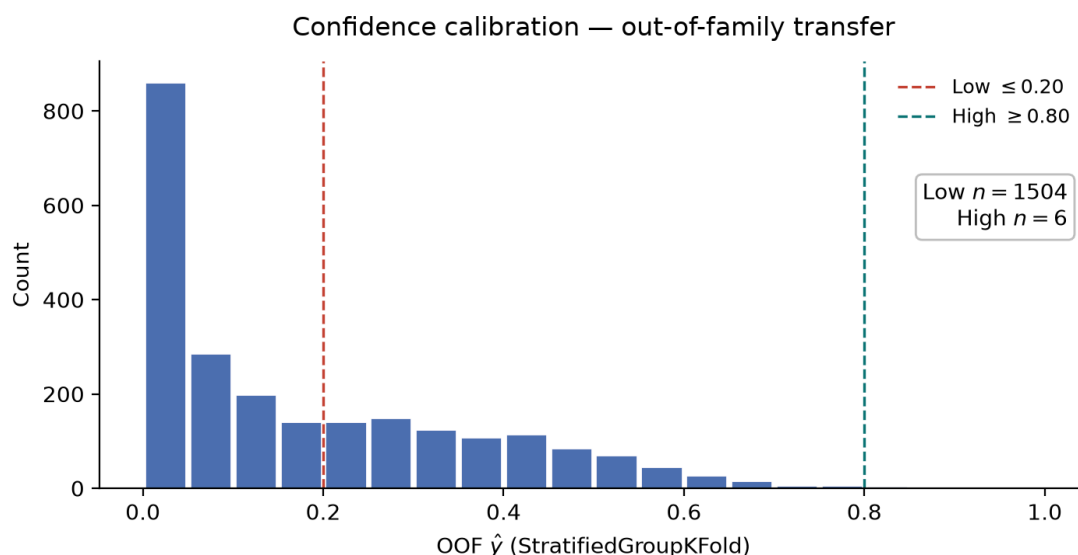


Figure 7 Out-of-fold prediction histogram of the RF model: Predictions for out-of-family sequences clustered below $\hat{y} \leq 0.20$ once family-specific k-mers were excluded.

4.2 EVA results

4.2.1 Generative Viability: EVA Generative Triage

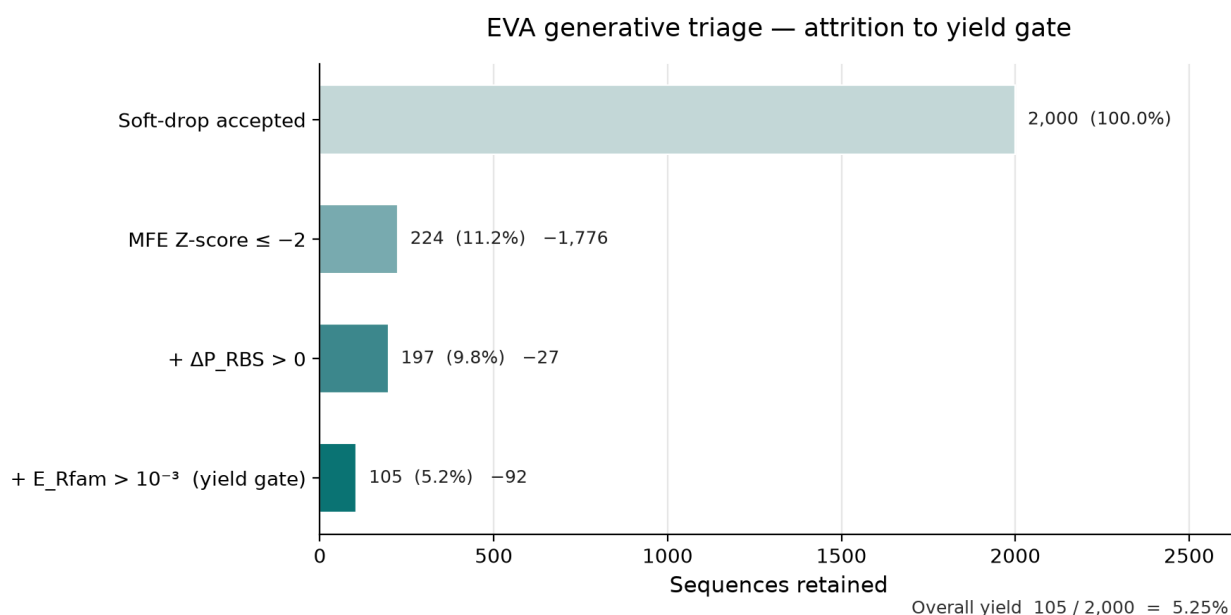


Figure 8 generative triage and biophysical attrition funnel: A filtering schematic that explains the attributable difference between genuine plausible thermoswitch-like structures from hallucinations. Candidates were first filtered for baseline structural stability (MFE Z-score ≤ -2), followed by the requirement for dynamic conformational shifting ($\Delta P_{RBS} > 0$), and finally gated for evolutionary viability ($E_{Rfam} > 10^{-3}$).

We filtered our 2,000 accepted generations through the three gates in **Section 3.3.3**. The set contracted to 105 sequences (5.25%, **Figure 8**). Comparative benchmarking across our candidate cohort confirmed that while ViennaRNA and NUPACK exhibit strong numerical agreement on static 37°C MFEs (Pearson $r > 0.95$), cross-engine concordance degraded substantially when computing dynamic melting temperatures and continuous thermal partition functions (Spearman $\rho \approx 0$). Consistent with broader discrepancies observed across RNA simulation software (Runge *et al.*, 2024), We designated ViennaRNA as the standardised baseline engine to eliminate cross-software calibration artefacts (**supplementary**).

We applied the baseline stability gate ($Z \leq -2$), dynamic unzipping filter ($\Delta P_{RBS} > 0$), and sequence novelty threshold ($E\text{-value} > 10^{-3}$) which resulted in marked attrition across the generative pool. In total, $N = 105$ candidates satisfied all three criteria simultaneously, representing an overall generative yield of 5.25% (**Figure 8**) advanced to continuous thermodynamic ensemble validation.

4.2.2 Thermal Validation: *In Silico* Thermal Sweeps

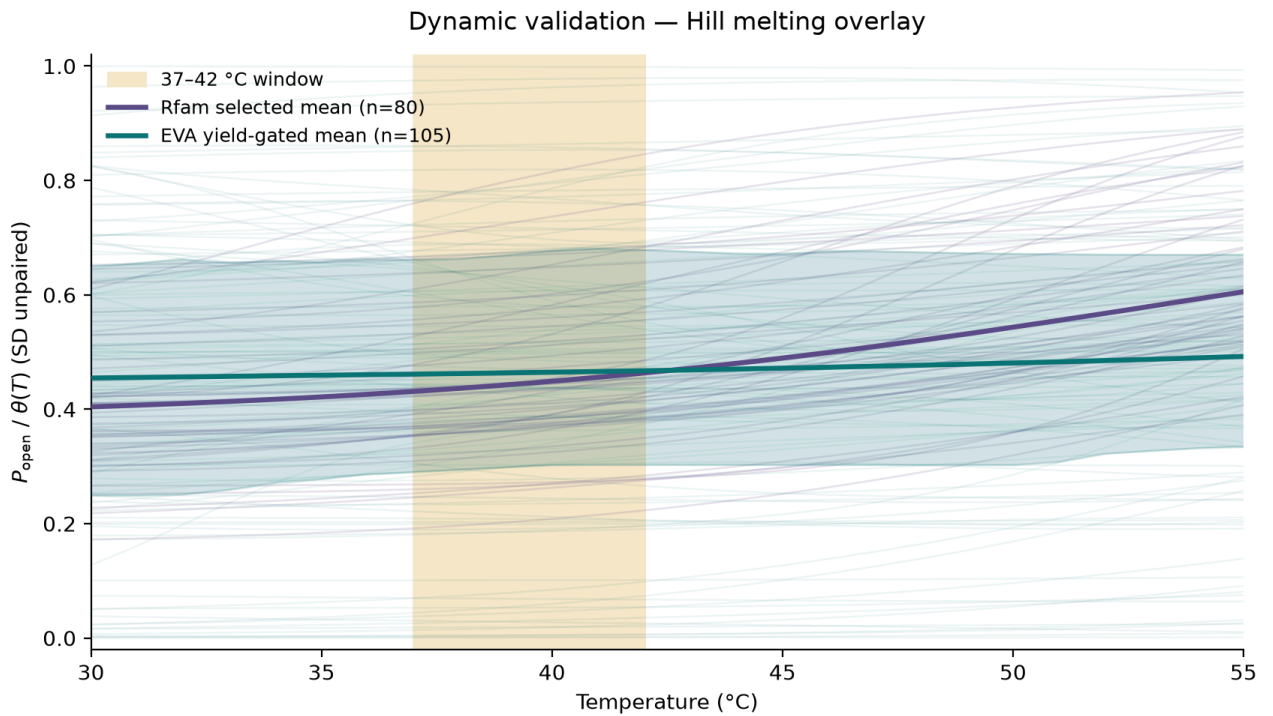


Figure 9 Dynamic thermal validation and Hill melting overlay: Dynamic thermal validation and Hill melting trajectory overlay across de novo candidates and natural controls. Solid lines denote cohort means shaded regions indicate the interquartile range (IQR, 25th-75th percentiles) of translation initiation unpairing probability (P_{open}) across temperatures. The natural control distribution represents a representative stratified subset of verified Rfam switches ($n = 80$) benchmarked against the yield-gated EVA cohort ($n = 105$).

Continuous thermodynamic melting profiles across the target window (37°C to 42°C) highlights a clear divergence between the generative cohort and natural switches (**Figure 9; Table 4**). While representative biological controls ($n = 80$) demonstrate a progressive thermal unzipping trajectory, the yield-gated EVA cohort ($N = 105$) exhibited elevated baseline folding stability ($\Delta G_{\text{density}} = -0.407 \text{ kcal} \cdot \text{mol}^{-1} \cdot \text{nt}^{-1}$) alongside a largely unresponsive opening profile across the continuous sweep ($\Delta P_{\text{RBS}} = 0.043$; **Table 4**), reflecting severe thermodynamic inertia across unguided generations (Huang *et al.*, 2026).

However, two top-ranking Tier-1 candidates diverged markedly from this unguided cohort baseline (**Table 4**). Candidate A (eva_sample_315) paired a tightly repressed baseline at 37°C with substantial dynamic unzipping and sharp cooperativity, transitioning with an activation threshold moderately above the targeted 42°C induction setpoint (Neupert *et al.*, 2008; **Table 4**). Candidate B (eva_sample_1858)

similarly demonstrated intermediate switching capacity with an activation threshold closely matching physiological induction. These isolates suggests that while average unguided generations remain structurally rigid, downstream biophysical screening successfully captures rare *de novo* candidates exhibiting dynamic, switch-like phase transitions approaching natural regulatory systems (Kortmann and Narberhaus, 2012).

Cohort / Candidate	Count (N)	Mean $\Delta G_{\text{density}}$ (kcal/mol/nt)	Mean $P_{\text{open}}(37^{\circ}\text{C})$ (ymin)	Mean $P_{\text{open}}(42^{\circ}\text{C})$	Mean ΔP_{RBS}	Activation T_m ($^{\circ}\text{C}$)	Hill Coeff. (h)
Natural Rfam Positive Controls	1198	-0.286	0.394	0.405	0.062	62.5	9.51
EVA Yield-Gated Cohort (All)	105	-0.407	0.461	0.467	0.043	53.7	10.20
EVA Tier-1 Candidate A (eva_sample_315)	1	-0.210	0.300	0.374	0.122	46.1	8.08
EVA Tier-1 Candidate B (eva_sample_1858)	1	-0.364	0.328	0.362	0.073	53.2	4.29

Table 4: Comparative thermodynamic and Hill parameters across natural controls, the full generative cohort, and Tier-1 hits. Summary of ViennaRNA ensemble thermodynamics and non-linear Hill equation fits across natural Rfam positive controls ($N = 1,198$), the yield-gated EVA cohort ($N = 105$), and the two top-performing Tier-1 candidates (eva_sample_315 and eva_sample_1858). Metrics include: thermodynamic density ($\Delta G_{\text{density}} = \Delta G_{\text{MFE}}/N$ at 37°C), baseline RBS accessibility at 37°C ($P_{\text{open}}(37^{\circ}\text{C})$), net dynamic unzipping ($\Delta P_{\text{RBS}} = P_{\text{open}}(55^{\circ}\text{C}) - P_{\text{open}}(37^{\circ}\text{C})$), activation threshold (T_m), and Hill cooperativity coefficient (h)

5. Discussion

Our current screening pipeline relies on thermodynamic engines that evaluate RNA at a static, global equilibrium. *In vivo*, however, RNA folding never settles into a single static state. It is an active, ongoing process that unfolds step-by-step as the transcript emerges from the RNA polymerase elongation complex (Mathews, 2004; Twittenhoff et al., 2020). By focusing solely on final, lowest-energy ensembles (Lorenz et al., 2011), our current equilibrium-based approach inadvertently favours over-stabilised

structures that remain kinetically trapped during temperature sweeps. Future iterations generally should incorporate kinetic folding simulators, such as Kinfold (Flamm et al., 2000) or RNAkinetics (Danilova et al., 2006). This will allow the scoring loop to evaluate transient intermediates and actively reward meta-stable, switchable conformations.

Planar 2D representations struggle to capture the complex, 3D energetics that drive cooperative melting (Tinoco and Bustamante, 1999; Cruz and Westhof, 2009). In nature, bacterial RNA thermometers rely heavily on tertiary interactions, non-canonical base pairing, and loop flexibility to execute sharp thermal transitions (Kortmann and Narberhaus, 2012; Feng et al., 2026). To capture these physics, recent frameworks combine energy-aware 3D generative models (like RNA-EFM) with atomic flexibility predictors (like DeepRMSF) to balance geometric realism with thermodynamic stability (Abir and Zhang, 2025; Feng et al., 2026). This directly counters the structural rigidity we observed across our generated designs. Ultimately, pairing future generative pipelines with true 3D structural predictors like RoseTTAFold2NA may provide the geometric feedback needed to achieve stable thermal switching in *de novo* candidates (Baek et al., 2024; Joshi et al., 2025).

Generative biological foundation models propose sequence candidates, but orthogonal filtering gates decide viability as further discussed (in the **supplementary**) with a similar *in silico* process. Unconstrained autoregressive sampling inherently encounters biological hallucination and functional inertia. Whether gated through multi-stage *in silico* biophysical screens ($Z \leq -2$, $\Delta P_{\text{RBS}} > 0$, $E_{\text{Rfam}} > 10^{-3}$) or wet-lab cellular plaque assays, rigorous downstream triage is essential to isolate the $\approx 5\%$ of generations that are genuinely functional.

Finally, while foundational models such as EVA effectively internalise the baseline sequence distributions of natural RNAs, unguided sampling remains computationally prohibitive and biophysically unconstrained. In our screening protocol, generating a viable yield of approximately 100 structural candidates required 2/3 days of continuous generative sampling due to extreme filter attrition ($\sim 95\%$ rejection). Although categorical taxonomy and RNA-type conditioning were utilised during sampling as recommended by Huang *et al.* (2026), residual thermodynamic inertia persists as a fundamental bottleneck of unguided autoregressive sequence generation. To transition from broad statistical imitation to directed biophysical engineering, the generative architecture could be formulated as a Reinforcement Learning (RL) policy. Unconstrained RL often underperforms without biophysical grounding (Runge *et al.*, 2024; Tang *et al.*, 2026). The policy must be guided by an explicit multi-state reward function designed to score the continuous 37-42°C unzipping delta rather than a static single-temperature objective.

6. References

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